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Class: CS-F

Roll no. 23i-0579

Home Work 24

Problem 1

$$A = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 0 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix} \xrightarrow{R_3 - 3R_1}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix} \xrightarrow{R_2 \times \frac{1}{2}} \begin{bmatrix} 1 & 0 & -2 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 - R_2 \\ R_3 - R_3 \end{matrix}}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 \times \frac{1}{4} \\ R_1 + R_3 \end{matrix}}$$

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_4 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, E_6 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}$$

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$$E_7 =$$

$$E_6 \times E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

 $E_6 \times E_5$ E_4

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1/4 & 1/4 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1/4 & 1/4 \end{bmatrix}$$

 $E_6 \times E_5 \times E_4$ E_3 E_2

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1/4 & 1/4 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1/2 & 0 \\ 0 & -1/8 & 1/4 \end{bmatrix}$$

 $E_6 \times E_5 \times E_4 \times E_3$ E_2

$$= \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1/2 & 0 \\ 0 & -1/8 & 1/4 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & 1 & 0 \\ 1/2 & 0 & 0 \\ -1/8 & -3/4 & 1/4 \end{bmatrix}$$

 $E_6 \times E_5 \times E_4 \times E_3 \times E_2$ E_1

$$EA = \left[\begin{array}{ccc|ccc} -1/2 & 1 & 0 & 0 & 2 & 4 & 0 \\ 1/2 & 0 & 0 & 1 & 1 & 0 & -1 \\ -1/8 & -3/4 & 1/4 & 3 & 4 & 2 & 1 \end{array} \right]$$

$$= \left[\begin{array}{cccc} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] = R$$

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Problem 2 Alternative method of problem 1

$$\left[\begin{array}{cccc|ccc} 9 & 2 & 4 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 \end{array} \right] R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 4 & 0 & -3 & 1 \end{array} \right] R_3 - 3R_1$$

$$\left[\begin{array}{cccc|ccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2 & 4 & 0 & -3 & 1 \end{array} \right] R_2 \times \frac{1}{2}$$

$$\left[\begin{array}{cccc|ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1 & -3 & 1 \end{array} \right] \begin{array}{l} R_3 - R_2 \\ R_1 - R_2 \end{array}$$

$$\left[\begin{array}{cccc|ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] R_3 \times \frac{1}{4}$$

$$R = \left[\begin{array}{ccc} -1/2 & 1 & 0 \\ 1/2 & 0 & 0 \\ -1/8 & -3/4 & 1/4 \end{array} \right]$$

Problem 3

$$a) \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$|A| = 3$$

$$\text{Adj } A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj } A}{|A|} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\det = 1, \text{Adj} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj } A}{|A|} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$c) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$|A| = -1$$

$$C_{11} = -1, C_{12} = 0, C_{13} = 0, C_{21} = 0$$

$$C_{22} = 0, C_{23} = -1, C_{31} = 0, C_{32} = -1, C_{33} = 0$$

$$\text{Adj } A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}^T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

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$$d) \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad |A| = 1$$

$$C_{11} = 1, C_{12} = 5, C_{13} = 0, C_{21} = 0, C_{22} = 1$$

$$C_{33} = 0, C_{31} = 0, C_{32} = 0, C_{33} = -1$$

$$\text{Adj } A = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\text{Prob 1 } A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\text{Problem 4} \quad \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}$$

A B AB

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$\left\{ \begin{array}{l} \\ \end{array} \right. = \begin{bmatrix} 1 & -2 \\ 0 & 1/3 \end{bmatrix}$$

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Problem 5

$$a) \begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}$$

$$|A| = 10 - 8 = 2$$

$$\text{Adj } A = \begin{bmatrix} 5 & -2 \\ -4 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 5/2 & -1 \\ -2 & 1 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix}$$

$$|A| = 6 - 6 = 0$$

Inverse not possible.

$$c) \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$|A| = 45 + 96 + 84 - 105 - 48 - 72 = 0$$

inverse not possible.

$$d) \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$$

$$|A| = 18 + 3 + 4 - 2 - 12 - 9 = 2$$

$$C_{11} = 18 - 12 = 6, \quad C_{12} = -5, \quad C_{13} = 1$$

e

$$| \quad 0 \quad 1 \quad \wedge \quad |$$

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$$C_{21} = -6, \quad C_{22} = 8, \quad C_{23} = -2$$

$$C_{31} = 2, \quad C_{32} = -3, \quad C_{33} = 1$$

$$\text{Adj } A = \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}^T = \begin{bmatrix} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix}$$